UNDERGRADUATE THESIS DEFENSE SEMINAR

When Does Federated BatchNorm Help under Differential Privacy?

An Eval-Regime Characterization on Medical ECG Classification

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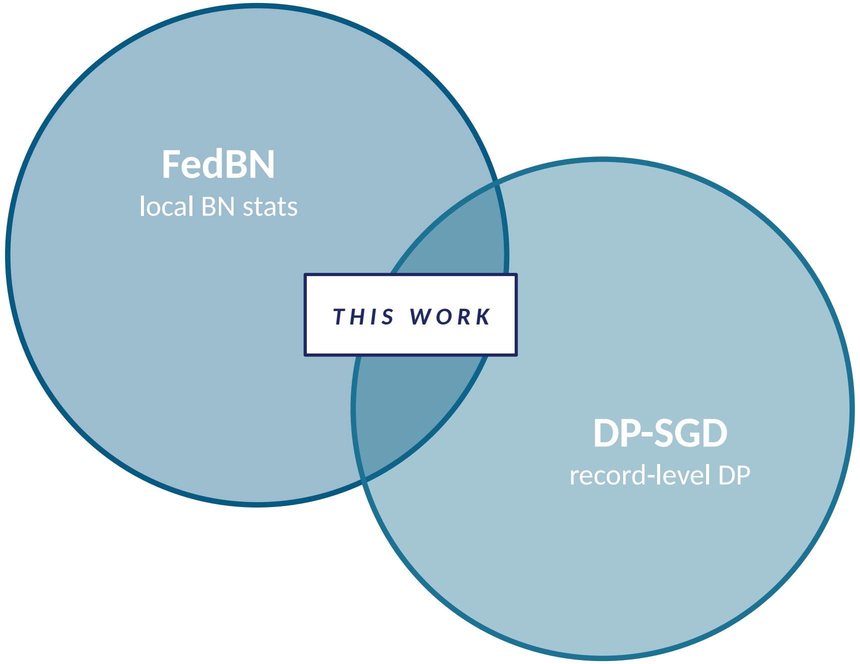
Motivation

Medical AI must reconcile two competing demands:

Train clinically meaningful models across hospital sites without centralizing patient data (Federated Learning), and provide formal per-record privacy guarantees during training (Differential Privacy).

The tension:

FedBN (keeps each site's BatchNorm statistics local) and DP-SGD (clips per-sample gradients) do not compose naturally — they collide at the library level (Opacus rejects BatchNorm modules outright).



Goal: Systematically map the boundary conditions of this composition on a real medical task (ECG classification).

Background · Why BatchNorm Breaks DP-SGD

DP-SGD requires per-sample gradients

Each training sample's gradient is clipped individually before noise is added, providing record-level (ε, δ)-DP.

BatchNorm gradients are not well-defined per sample

BN normalizes each sample using the batch's mean and variance — i.e., each sample's output depends on the other samples in the batch.

Opacus enforces this constraint

The library's ModuleValidator throws UnsupportedModuleError on any BatchNormNd layer at initialization, halting the privacy engine setup.

opacus_error.py

```
from opacus import PrivacyEngine

model = CNN1D()  # has BatchNorm1d layers
engine = PrivacyEngine()
model, opt, dl = engine.make_private(...)

UnsupportedModuleError:
Model contains BatchNormNd layer
which is not supported.

# Standard workaround: replace BN
# with GroupNorm — but that breaks
# FedBN's personalization mechanism.
```

Our Construction · DP-FedBN Dual-Optimizer

Key observation:

Under the FedBN protocol, BN parameters never cross the federation boundary — so they need no DP protection in this threat model.

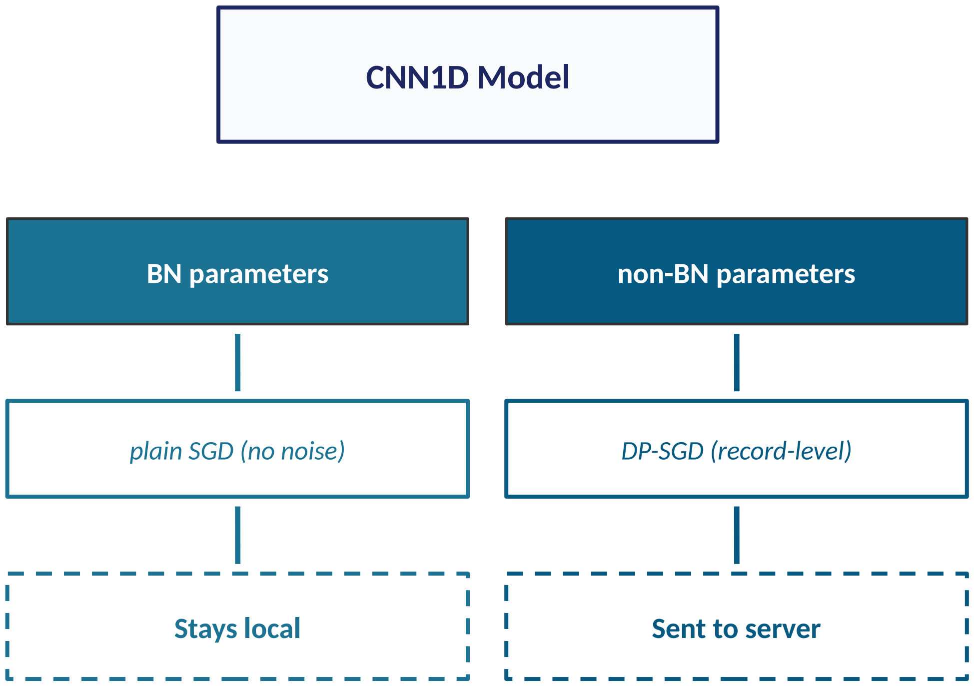
Construction (four steps):

1. Freeze BN parameters before privacy-engine wrap
2. Opacus's ModuleValidator now accepts the model (BN params have no grad)
3. Unfreeze BN; create two separate optimizers — DP-SGD for non-BN, plain SGD for BN
4. Run FedBN protocol: BN parameters stay local; only non-BN parameters enter aggregation

Result:

Off-the-shelf Opacus 1.5 + record-level DP-SGD + preserved FedBN personalization, all in one model.

Model Parameter Flow



Contributions

1

Methodological

DP-FedBN dual-optimizer

Off-the-shelf Opacus pipeline via freeze-then-unfreeze; record-level DP-SGD composed with FedBN's BN-locality. The term 'DP-FedBN' appears in ADCOL/ICML 2023 as a one-paragraph party-level Laplace baseline — we give the first record-level construction.

2

Empirical Finding #1

Eval-regime flip

FedBN's ranking inverts systematically between pooled-central and per-client local evaluation regimes (+0.72 F1 swing on MIT-BIH label-skew). Confirmed on both medical tasks (MIT-BIH + PTB-XL). Task-agnostic.

3

Empirical Finding #2

Partition-dependent advantage

DP-FedBN's local advantage hinges on partition type: under label-skew it gains +0.17 F1 (ε=3) over the GroupNorm fallback; under Dirichlet skew the advantage flips to negative. Yields a heterogeneity-typed decision rubric for practitioners.

Broader Impact

Academic

Targeted submission to IEEE Journal of Biomedical and Health Informatics (JBHI), September 2026. First systematic characterization of the FedBN + DP composition in the medical-FL literature.

~40

experimental configs

Practitioners

Directly actionable decision rubric for medical-FL practitioners: when to choose DP-FedBN versus the GroupNorm fallback, conditional on partition heterogeneity and privacy budget.

2

medical tasks (MIT-BIH + PTB-XL)

Engineering

Reproducible, Opacus 1.5-compatible reference implementation. Concrete resolution for the two-year-old engineering bottleneck documented in Opacus GitHub Issue #472.

Open source

(upon acceptance)

Feasibility

Compatible with off-the-shelf tooling

Opacus 1.5 + PyTorch 2.4 + custom FL simulator — no forks, no patches; suitable for production deployment.

Bit-exact reproducibility

All configurations, logs, and seed (=42) preserved; cuDNN deterministic mode for run-to-run reproducibility.

Clinically appropriate data handling

AAMI EC57:2012-compliant labeling; patient-stratified partitioning reflecting real non-IID hospital cohorts.

Modest hardware requirements

Full sweep (~40 configs) runs in ~24 hours on a single GTX 1660 Super GPU; portable to higher-end hardware.

TECHNOLOGY STACK

Language	`Python 3.11`
Deep learning	`PyTorch 2.4 + CUDA`
Privacy	`Opacus 1.5`
FL engine	`Custom sequential sim`
Data	`MIT-BIH, PTB-XL`
Analysis	`pandas, NumPy, SciPy`
Plots	`Matplotlib + LaTeX`
Versioning	`Git + GitHub`

Methodology · Data and Experimental Setup

MIT-BIH Arrhythmia

Records	**48 patients**
Beats	**~109,453**
Classes	**5 (AAMI EC57)**
Task	**Beat classification**

PTB-XL

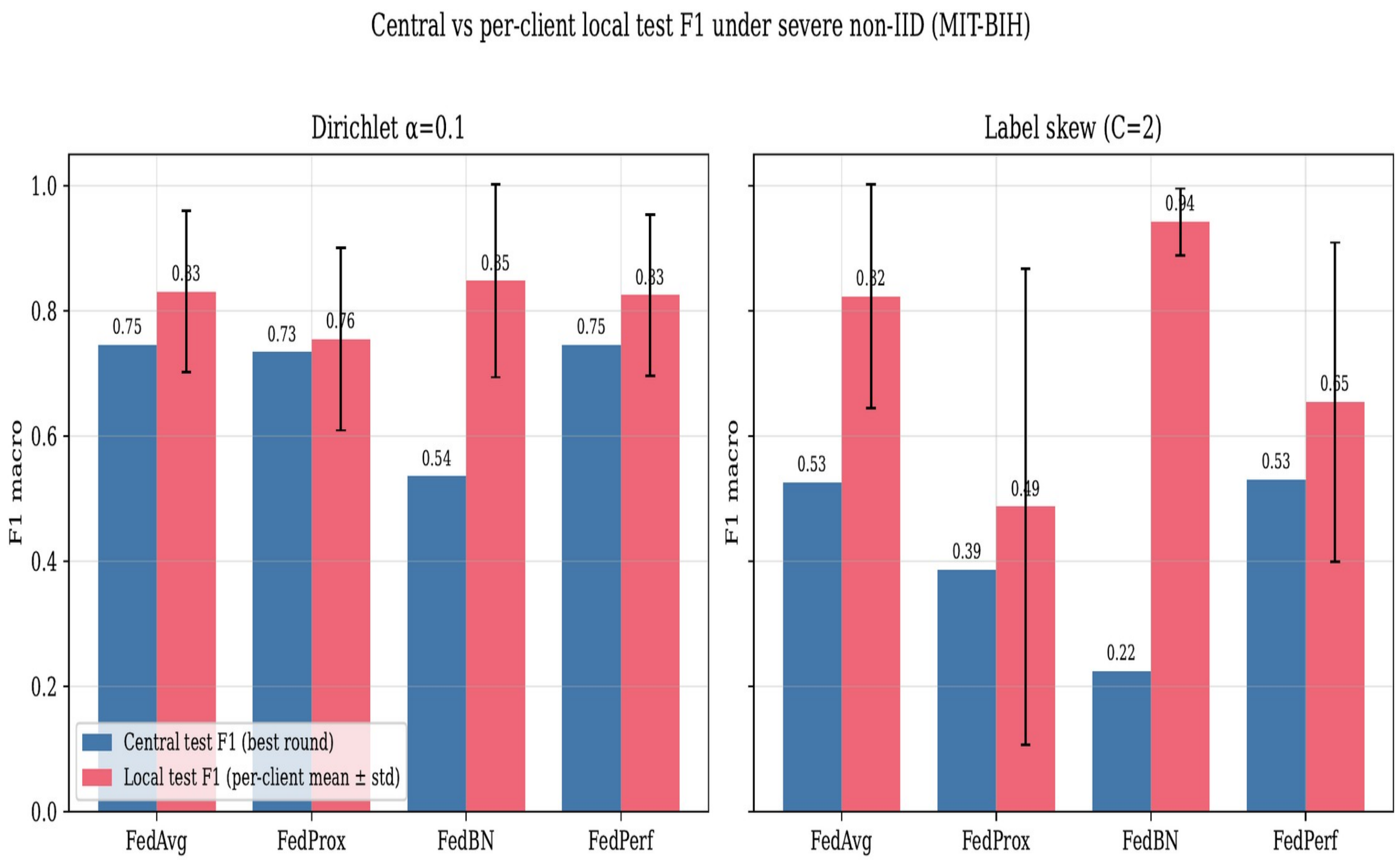
Records	**21,388**
Length	**10 sec / record**
Classes	**2 (binary)**
Task	**Record-level classification**

Heterogeneity Regimes (5 clients)

`iid`	IID baseline
`quantity_skew`	Imbalanced data volumes
`label_skew_c2`	2 classes/client
`label_skew_c3`	3 classes/client
`dirichlet_a01`	Dirichlet α=0.1 (severe)
`dirichlet_a05`	Dirichlet α=0.5 (moderate)

~40	6	3	4	5
configurations	partition regimes	DP budgets (ε=1, 3, 10)	FL aggregation methods	clients × 30-50 rounds

Results · The Eval-Regime Flip



MIT-BIH across 6 partition regimes: pooled-central F1 (left bars) vs per-client local F1 (right bars).

LARGEST SWING

+0.72

FedBN · label_skew_c2

Pooled F1: 0.224 → Local F1: 0.942

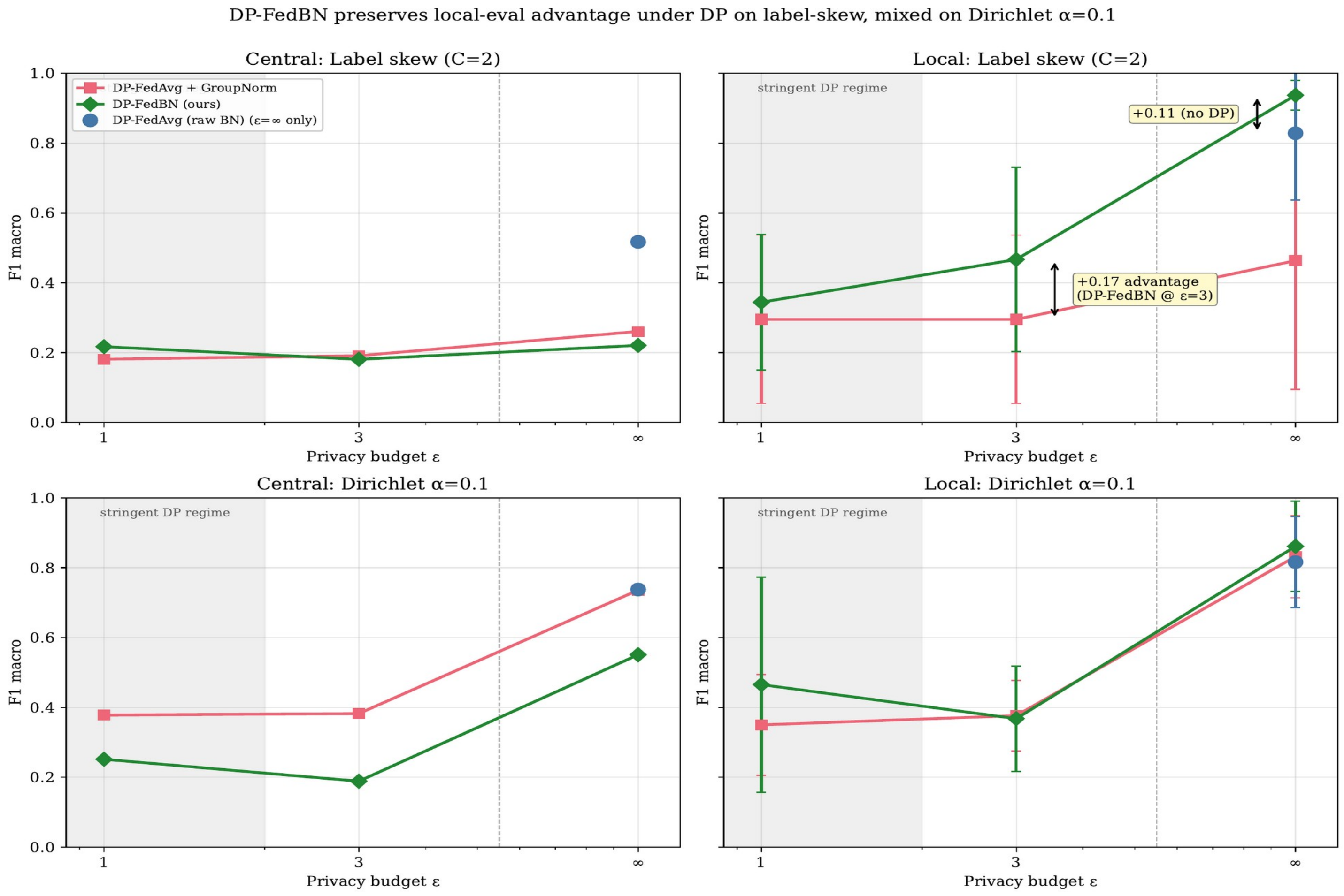
Finding

FedBN's ranking inverts across evaluation regimes: worst under pooled-central, best under per-client local.

Practical implication

Medical-FL benchmarks must explicitly state which evaluation regime is reported. Both should be reported when deployment heterogeneity is unclear.

Results · Privacy-Utility Tradeoff



Macro F1 vs DP budget ε: DP-FedBN (local eval) vs DP-FedAvg+GN (central eval), MIT-BIH.

KEY FINDINGS

1. Under label-skew: DP-FedBN beats the GN fallback by +0.17 local F1 at ε=3.

2. Under Dirichlet skew: the advantage flips negative (−0.234). Partition type is determinative.

3. Cross-task on PTB-XL: same pattern replicates. Conclusion is not dataset-specific.

PRACTITIONER RUBRIC

If label-truncated cohorts + local deploy
→ use DP-FedBN

If Dirichlet-style class-mix heterogeneity
→ use DP-FedAvg + GroupNorm

Limitations, Roadmap, and Closing

LIMITATIONS

Single-seed (=42)

Variance unquantified; multi-seed expansion in progress.

CNN1D only

Architecture-dependent generalization untested.

5 clients

Production medical FL has tens to hundreds.

FedProx tuning

μ=0.01 likely under-tuned.

ROADMAP (MAY-SEP 2026)

Multi-seed (3-5)

PTB-XL Dirichlet priority; statistical CI bands.

FedProx grid

$\mu \in \{0.001, 0.01, 0.1, 1.0\}$.

Threat model

BN repeated-query attack analysis.

Ablations

With/without freeze trick comparison.

TARGET

IEEE JBHI

Journal of Biomedical and Health Informatics.

September 2026

Submission deadline.

First-round review

Major revision expected; revisions planned.

Backup venues

ML4H workshop · MICCAI MedAGI.

Thank you · Questions and discussion welcome · **efedenizbaglar45@gmail.com**